Deep-Learning MNIST Digit Classifier

1st Aakanksha Rawat
MEng in Computer Vision and AI
University of Limerick
Ireland
25243829@studentmail.ul.ie

2nd Aga Asghar Murthuza
MEng in Computer Vision and AI
University of Limerick
Ireland
25210645@studentmail.ul.ie

3rd Bhagyalakshmy Saburaj
MEng in Computer Vision and AI
University of Limerick
Ireland
25275224@studentmail.ul.ie

4th Rosanne Fernando
MEng in Computer Vision and AI
University of Limerick
Ireland
25275216@studentmail.ul.ie

5th Roshni Ravi
MEng in Computer Vision and AI
University of Limerick
Ireland
25015168@studentmail.ul.ie

***Abstract*—Handwritten digit classification remains a fundamental problem in machine learning and computer vision. While traditional methods relied on manually engineered features and simple classifiers, limiting their robustness to varied handwriting styles and noise, this project focuses on developing an accurate and efficient automated digit recognition system. Utilizing the MNIST dataset, we benchmark the performance of three modern learning algorithms: Support Vector Machine (SVM), Generalized Learning Vector Quantization (GLVQ), and a Convolutional Neural Network (CNN). The CNN model, implemented via MatConvNet, is highlighted for its superior ability to perform automatic feature extraction and achieve high accuracy with efficient CPU-based training, resulting in reduced computation time. The developed approach not only achieves reliable digit recognition but also demonstrates potential for extension to general handwriting and alphabet recognition. This work underscores the practical benefits of deep learning and computer vision in enhancing automation, reliability, and efficiency across various real-world applications, including postal automation, bank check processing, and large-scale form digitization.**

***Index Terms*—Convolutional Neural Network, Digit Recognition, MNIST dataset, Computer Vision, Classification Accuracy**

I. Introduction

Convolutional Neural Networks (CNNs) have emerged as one of the most powerful and widely used approaches in machine learning and computer vision. They have been instrumental in achieving remarkable success in challenging applications such as object detection, image segmentation, and face recognition, particularly demonstrated in competitions like ImageNet.

Motivated by these advancements, we adopt CNNs for our image classification task. Specifically, we apply this approach to handwritten digit recognition, a problem of both academic significance and practical importance. Handwritten digit recognition has numerous real-world applications for instance, automating bank check processing, sorting postal mail, and digit recognition in various administrative and business forms. By leveraging CNNs, we aim to build an efficient and accurate system capable of addressing these real-life challenges effectively.

A. History

The study of handwritten digit classification has a long history, dating back to the early stages of computer vision and pattern recognition. In the 1960s, researchers began developing methods to automate the recognition of handwritten characters to reduce the workload of manual data entry. One of the first notable systems was created by David H. Shepard in 1968, which used template matching to identify handwritten digits. However, these early systems struggled with variations in handwriting and required heavy image preprocessing.

During the 1980s and 1990s, improvements in machine learning algorithms and computing power led to more advanced techniques for handwritten digit classification. The introduction of neural networks, especially the Multi-Layer Perceptron (MLP), was a key breakthrough, as it allowed models to learn features directly from raw pixel data. A major milestone came in 1998 with LeNet-5, developed by Yann LeCun and his team. It was one of the first Convolutional Neural Networks (CNNs) designed specifically for recognizing handwritten digits and achieved impressive results on the MNIST dataset, laying the groundwork for modern deep learning systems.

In the early 2000s, interest shifted toward ensemble methods and Support Vector Machines (SVMs) to further boost classification accuracy. Techniques such as bagging and boosting were used to combine multiple models for better performance, while SVMs became popular for their effectiveness in finding optimal decision boundaries in complex, high-dimensional data. Researchers also worked extensively on improving feature engineering, which ultimately helped drive the rise of deep learning methods in later years.

B. Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are a type of deep learning model designed to work with visual data. They are commonly used for image classification, image grouping, and object detection in complex scenes. CNNs have achieved outstanding results in many practical applications, including face and object recognition, reading street signs, analyzing medical images, and other tasks involving visual pattern detection.

The core component of a CNN is the convolutional layer. It contains a set of filters (also called kernels) that move across the image to detect specific features. Each filter focuses on a small area of the image but processes all its depth channels. As the filter slides over the image, it performs mathematical operations to create feature maps that highlight patterns such as edges, textures, and shapes.

These feature maps are then passed through pooling layers, which reduce their size by keeping only the most important information. This step helps the network become faster and more robust to small changes or shifts in the image. Finally, one or more fully connected layers combine the extracted features and assign the image to a specific class, producing a final predicted label.

II. Related Works

In the paper "Embarking on the MNIST Handwritten Digit Classification Challenge" by Dr. S.K. Mahboob, several methods for handwritten digit recognition were reviewed using the MNIST dataset. The study highlights that Convolutional Neural Networks (CNNs) achieved over 99% accuracy and became the foundation of modern image classification. Other approaches such as Reservoir Computing, DCT-based ANN, LIRA, Optimum Path Forest, and CNN–SVM hybrids were also discussed. [1].

The study "Comparisons of Deep Learning Algorithms for MNIST in Real-Time Environment" compares MLR, CNN, ResNet-34, and CapsNet for handwritten digit recognition. Using TensorFlow and a Java-based GUI, performance was evaluated on accuracy, speed, and memory. CapsNet achieved the best results with 99.4% accuracy, while ResNet and CNN showed moderate accuracy and slower runtimes. MLR was fastest to train but least accurate. [2].

Shengjie Zhai proposed a CNN-based model for handwritten digit recognition on the MNIST dataset with optimized hyperparameters. The network includes three convolutional and two fully connected layers. By tuning batch size, kernel size, and learning rate, the model achieved 99.82% training and 99.40% testing accuracy. They proved that hyperparameter optimization is vital for high accuracy in digit recognition [3].

The paper "MNIST Handwritten Digit Recognition Using a Deep Learning-Based Modified Dual Input CNN (DICNN) Model" proposes a dual-input CNN for digit recognition on the MNIST dataset. Two parallel CNN streams with ReLU and MaxPooling layers are merged and passed through dense layers for classification. Trained with the Adam optimizer, the model achieved 98.7% test accuracy, outperforming traditional models like KNN, SVM, and RF. [4].

Biswas proposed an efficient CNN model for handwritten digit recognition using MNIST and Kaggle datasets. The best design, with four convolutional and two fully connected layers, was trained using SGDM, outperforming Adam and RMSProp. It achieved 99.53% accuracy on MNIST and 98.93% on Kaggle. The study showed that simple CNN architectures can deliver high accuracy with lower computational cost [5].

In the paper "Handwritten Digits Recognition Using SVM, KNN, RF and Deep Learning Neural Networks" compares SVM, KNN, RF, MLP, and CNN for digit recognition on MNIST. Using OpenCV preprocessing and resizing to 28×28, CNN achieved the best accuracy 97.6% on industrial images and up to 100% on MNIST. While MLP improved slightly with more layers, CNN required longer training. The study concludes that effective preprocessing and image transformation greatly enhance recognition accuracy [6].

The author of the paper "Handwritten Digits Identification Using MNIST Database via Machine Learning Models" Birjit compared SVM, KNN, Random Forest, Naïve Bayes, and ANN for digit recognition in MNIST. SVM and Random Forest achieved the best accuracy (95–98%), showing that proper image preprocessing greatly improves performance. The study concluded that SVM-based models remain efficient for handwritten digit classification [7].

Yevhen Chychkarov compared SVM, KNN, Random Forest, and CNN models for handwritten digit recognition on the MNIST dataset. They found that CNN achieved the best accuracy, confirming the superiority of deep learning over traditional methods [8].

Rana in the paper "Comparison of the Error Rates of MNIST Datasets Using Different Types of Machine Learning Model" compared Logistic Regression, SVM, Random Forest, and CNN for handwritten digit recognition. CNN achieved the lowest error rate (1.2%), outperforming traditional models like SVM (8.2%) and Random Forest (3.7%), proving deep learning's superiority for MNIST classification [9].

III. Proposed Method

mThe proposed methodology focuses on developing an efficient deep learning-based model for handwritten digit classification using the MNIST dataset. The system uses a convolutional neural network (CNN) architecture to automatically extract spatial features from grayscale images of digits and perform accurate classification into ten distinct classes (0–9). To further enhance the performance of the model, a comparative analysis was conducted between two CNN architectures with varying depths and dropout configurations. The study emphasizes both accuracy and computational efficiency, with training and evaluation performed under consistent experimental settings to ensure fair comparison.

A. Data Collection and Preprocessing

The MNIST dataset, consisting of 60,000 training images and 10,000 grayscale test images of handwritten digits, was used to build the model. Each image has a spatial resolution of 28×28 pixels and a single intensity channel. Before training, all images were normalized to a range of [0, 1] by dividing pixel values by 255. This normalization step facilitates faster convergence and numerical stability during optimization. The dataset was reshaped as (28, 28, 1) to match the CNN input requirements. Labels were encoded as integers from 0 to 9, and the dataset was randomly shuffled to prevent class bias during training.

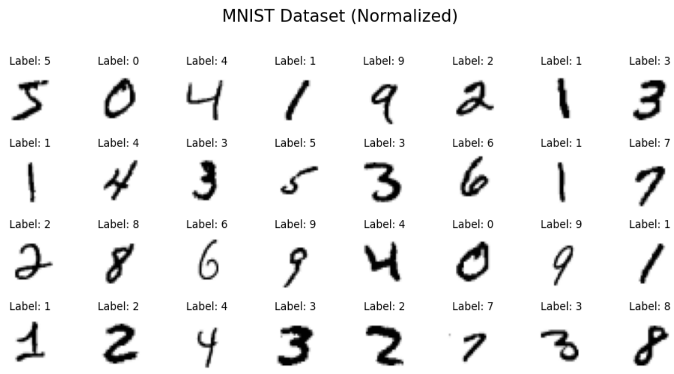

Fig. 1. A sample from MNIST dataset

B. Model Architecture

The proposed CNN model consists of two convolutional blocks followed by dense layers for classification.

- The input layer takes an image with a shape of (28, 28, 1), suggesting a 28x28 grayscale image.
- The first convolutional layer (conv2d-44) uses 32 filters and produces an output shape of (28, 28, 32). The dimensions remain 28x28 because the ”same” padding was used.
- A max-pooling operation (max-pooling2d-30) of size 2x2 follows, reducing the spatial dimensions to (14, 14) while keeping 32 feature maps.
- The second convolutional layer (conv2d-45) takes the (14, 14, 32) feature maps and applies 64 filters, increasing the depth to 64. The output shape is (14, 14, 64).
- A second max-pooling operation (max-pooling2d-31) of size 2x2 reduces the spatial dimensions further to (7, 7), resulting in (7, 7, 64).
- Another Dropout layer (dropout-34) is applied to the resulting feature maps.
- A third convolutional layer (conv2d-46) follows, utilizing 128 filters. The spatial dimensions are reduced to (5, 5), producing an output of (5, 5, 128).
- The 3D feature maps from the final convolutional layer are Flattened (flatten-15) into a 1D vector of size 5x5x128 =3200 features.
- This vector is fed into the first fully connected (Dense) layer (dense-29) consisting of 128 neurons.
- A third Dropout layer (dropout-35) is applied to the output of the 128-neuron dense layer for regularization.
- Finally, the classification is performed by the output Dense layer (dense-30), which has 10 units. This layer uses the softmax activation function to generate probability distributions over the ten digit classes.

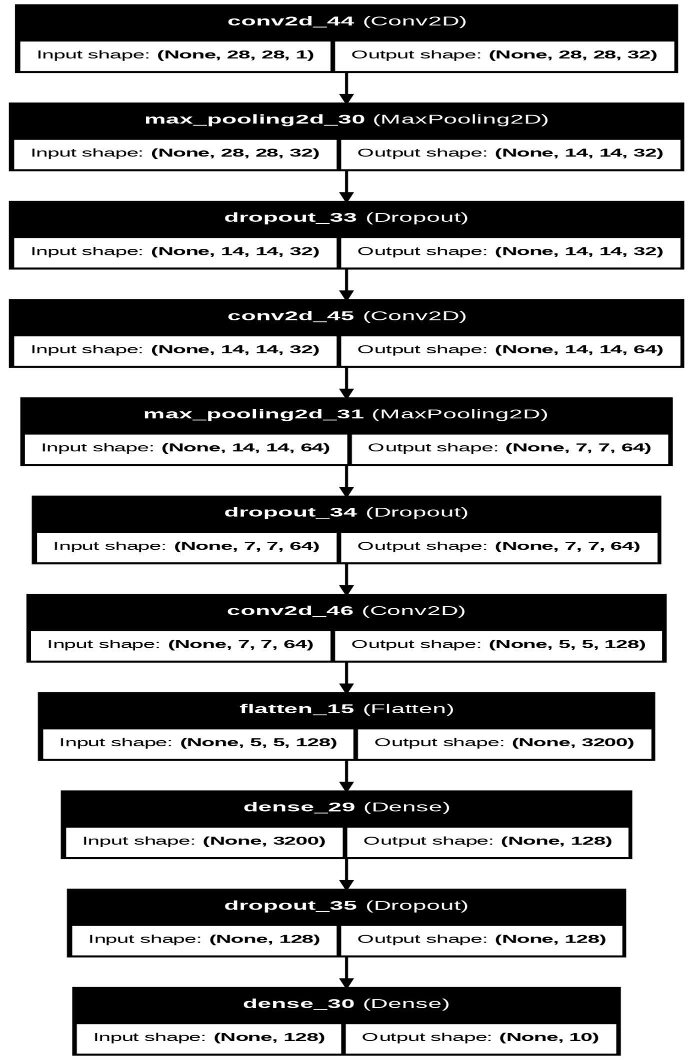

Fig. 2. Model Architecture of CNN using optimizer as RMSprop

C. Training Configuration

The model uses two convolutional blocks to progressively learn hierarchical features (edges to patterns) from the small 28x28 input, balancing complexity and efficiency. Three Dropout layers are strategically implemented to regularize the model and prevent overfitting during both feature extraction and final classification.The network was compiled using the Sparse Categorical Cross-Entropy loss function, suitable for multi-class classification with integer labels. Two different optimizers, RMSprop and Adam were used across experiments to study the effect of optimization on convergence. Both models were trained for 50 epochs with a batch size of 64. The training data was divided into mini-batches, and the validation performance was monitored after each epoch to prevent overfitting. Weight initialization was handled automatically, ensuring stable gradient propagation throughout the network.

IV. Results

A. Accuracy on different trial runs

To ensure the robustness of the models, three tests were carried out. Each test was performed with identical model architecture and hyperparameters to verify the consistency and reliability of the results.

TABLE I
Test Run accuracies of both the models

Model	Data	Accuracy (%)	Loss
Model-1	Training	99.23	0.0220
	Validation	99.37	0.0207
	Testing	99.19	0.0286
Model-2	Training	99.19	0.0380
	Validation	99.16	0.0389
	Testing	99.19	0.0286

These results indicate that the models perform consistently across different tests and configurations, surpassing the target accuracy of 99%. The model demonstrates high generalization ability, as evidenced by the small variation in performance across the trials.

B. Training and Validation Performance

The model-1, having optimizer as RMSprop was trained for 50 epochs. Training and validation accuracy and loss curves are shown in Figures 3

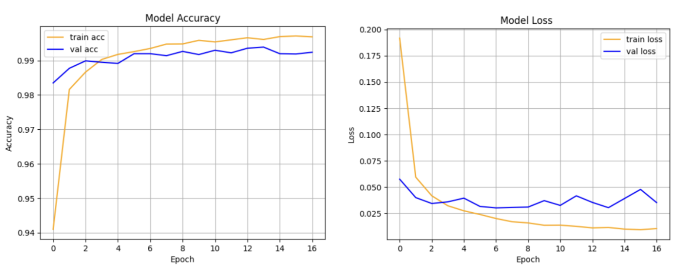

Fig. 3. Training vs. Validation Accuracy and Loss Graphs

The model-2, having optimizer as Adam was trained for 50 epochs. Training and validation accuracy and loss curves are shown in Figures 4

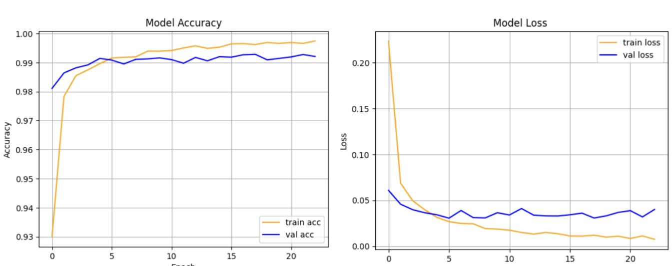

Fig. 4. Training vs. Validation Accuracy and Loss Graphs

V. Model Comparison

In order to evaluate the efficiency and robustness of the CNN model trained using the RMSprop optimizer, a comparative analysis was conducted with another CNN model optimized using Adam. The results of this comparative study are presented as follows.

A. Confusion Matrix Analysis

A detailed analysis of the confusion matrices, shown in Figures 5 and 6, highlights the classification consistency of both models.

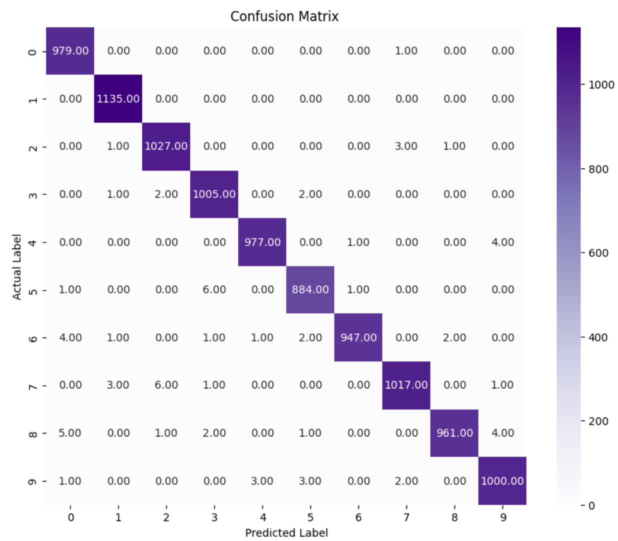

Fig. 5. Confusion matrix of the CNN model trained using RMSprop.

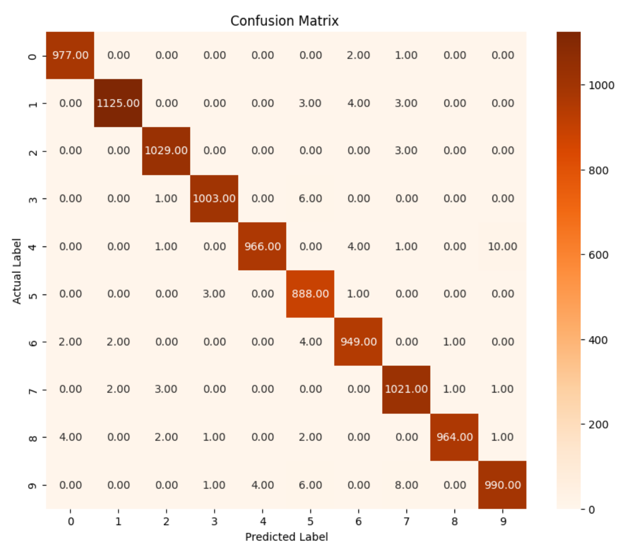

Fig. 6. Confusion matrix of the CNN model trained using Adam.

- Model-1 trained using RMSprop has Correct Classifications= 979 + 1135 + 1027 + 1005 + 977 + 884 + 947 + 1017 + 961 + 1000 = **9932**. Whereas, Model-2 trained using Adam has Correct Classification= 977 + 1125 + 1029 + 1003 + 966 + 888 + 949 + 1021 + 964 + 990 = **9912**.

- Model-1 has fewer significant misclassifications. For instance, the actual label '8' being predicted as '0' occurs 5 times. The most common error is the actual label '9' being predicted as '3' 3 times.
- Model-2 has slightly higher misclassification counts in certain areas. For the actual label '4', it is misclassified as '9' 10 times, which is a high number relative to other errors. The actual label '9' is misclassified as '7' 8 times.

V. CONCLUSION

This project successfully implemented a Convolutional Neural Network (CNN) for deep learning-based MNIST digit classification, achieving a testing accuracy of 99.29%. The model's high performance was driven by:

1) The 3×3 kernels effectively captured fine-grained spatial features, contributing to improved classification accuracy.
2) Dropout (20%) and early stopping mechanisms mitigated overfitting.
3) The model demonstrated robustness and stability, yielding consistent results.

The model-1 having RMSprop as optimizer, achieved a slightly higher overall accuracy of 99.37% and a lower loss value (0.0207) compared to model-2 having Adam as optimizer (99.19% accuracy, 0.0286 loss). Confusion matrices also revealed that the RMSprop model produced marginally cleaner class separations, exhibiting fewer off-diagonal errors and more uniform classification consistency across all digit classes.

VI. ACKNOWLEDGMENT

We would like to express our sincere gratitude to Prof Dr. Colin Flanagan for his valuable guidance, support, and insightful lectures throughout this project. His expertise and feedback were instrumental in the successful completion of this work.

REFERENCES

[1] Basha, SK Mahboob, et al. "EMBARKING ON THE MNIST HANDWRITTEN DIGIT CLASSIFICATION CHALLENGE: A BEGINNER'S JOURNEY INTO IMAGE DATA ANALYSIS." Journal of Science Engineering Technology and Management Science 2.06 (2025).

[2] Palvanov, Akmaljon, and Young Im Cho. "Comparisons of deep learning algorithms for MNIST in real-time environment." International Journal of Fuzzy Logic and Intelligent Systems 18.2 (2018): 126-134.

[3] Shao, Haijian, et al. "MNIST Handwritten Digit Classification Based on Convolutional Neural Network with Hyperparameter Optimization." Intelligent Automation & Soft Computing 36.3 (2023).

[4] M. Azgar, Ali, et al. "MNIST handwritten digit recognition using a deep learning-based modified dual input convolutional neural network (DICNN) model." International Congress on Information and Communication Technology. Singapore: Springer Nature Singapore, 2024.

[5] Biswas, Angona, and Md Saiful Islam. "An efficient CNN model for automated digital handwritten digit classification." Journal of Information Systems Engineering and Business Intelligence 7.1 (2021): 42-55.

[6] Chychkarov, Yevhen, et al. "Handwritten Digits Recognition Using SVM, KNN, RF and Deep Learning Neural Networks." CMIS 2864 (2021): 496-509.

[7] Gope, Birjit, et al. "Handwritten digits identification using mnist database via machine learning models." IOP conference series: materials science and engineering. Vol. 1022. No. 1. IOP Publishing, 2021.

[8] Rana, Md Suhel, Md Humayun Kabir, and Abdus Sobur. "Comparison of the error rates of MNIST datasets using different type of machine learning model." North Am Acad Res 6.5 (2023): 173-181.

[9] Chychkarov, Yevhen, et al. "Handwritten Digits Recognition Using SVM, KNN, RF and Deep Learning Neural Networks." CMIS 2864 (2021): 496-509.

[10] Bleed AI Academy, "Game with Body Gestures using Pose Detection",2022, https://youtu.be/Z2EGhplFOHs?feature=shared

[11] Human Pose Estimation dataset from Kaggle, https://www.kaggle.com/datasets/trainingdatapro/pose-estimation

[12] Sheryians Coding School, "Create 3D Animations Using HTML, CSS & JS — Scrolling Animation Using Canvas", 2023, https://youtu.be/Ud_hP2raTmk?si=4HtjBx9yVsVHgoHK